Estimating in Building Construction

Chapter 10 Concrete

Estimating Concrete

- Separate concrete by:
  - Type
  - Strength
  - Color
  - Special testing and curing

Concrete Footings

- Found on
  - Wall sections
  - Foundation plan
- Look for different sizes under different walls

Concrete Footings

Volume = Length x Width x Depth

- Measured in cubic yards

Example 10.1 Continuous Footings

See Fig 10.1 Fig 10.2

Determine quantity of concrete in 3′ 2″ wide continuous footing

Total length of wall = 337.333′

Cross-section = 3′ 2″ x 1′

= 3.1667 x 1= 3.167

Vol of concrete = 3.167 x 337.333

= 1068 cf

= 39.6 cy

Add 5% for waste

Vol of concrete = 39.6 x 1.05

= 42 cy

Side of Wall	Length
A	39′ 10″
B	34′ 4″
C	14′ 10″
D	21′ 10″
E	18′ 2″
F	30′ 8″
G	32′ 0″
H	6′ 10″
I	34′ 4″
J	6′ 10″
K	42′ 0″
L	55′ 8″
Total	337′ 4″

Example 10.2 Spread Footings

See Fig 10.4

Determine quantity of concrete

in 3′ x 3′ x 1′ spread footing

Vol of concrete = 3 x 3 x 1

= 9 cf

= 0.33 cy

Add 5% for waste

Vol of concrete = 0.33 x 1.05

= 0.35 cy

use 1 cy

Example 10.4 Formed Piers

See Fig 10.4

Determine quantity of concrete for formed pier 1′ diameter x 3′ 7″ tall

Vol of concrete = πr^2 x 3′ 7″

= 3.142 x 0.5 x 0.5 x 3.58

= 2.81 cf = 0.1 cy

Add 5% for waste

Vol of concrete = 0.1 x 1.05

= 0.12 cy

use 1 cy

Example 10.5 Foundation Wall

See Fig 10.8

Determine quantity of concrete

Building Perimeter wall = 1′ 2″ thick

Interior wall = 1′ thick

Vol of concrete = 303′ x 3′ 8″ x 1′ 2″

= 1301 cf = 48 cy

Wall "B" 8′ 4″ high

Vol of concrete = 32′ 4″ x 8′ 4″ x 1′ 2″

= 315 cf = 11.7 cy

Concrete in exterior foundation walls

= 48 + 11.7 = 59.7 cy

Add 5% for waste

Vol of concrete = 59.7 x 1.05

= 63 cy

Side of Wall	Length
A	38′ 10″
C	13′ 10″
D	23′ 10″
E	16′ 2″
F	32′ 8″
G	30′ 0″
H	8′ 10″
I	32′ 4″
J	8′ 10″
K	40′ 0″
L	57′ 8″
Total	303′ 0″

*Placing Concrete

Typical Productive Rates	
Type of Placement	LHR/c.y.
Continuous Footing - Direct Chute	0.4
Spread Footings – Direct Chute	0.873
Drilled Piers	0.320
Formed Piers	0.873
Foundation Walls – Direct Chute	0.5
Grade Beams	0.4
Slab – Direct Chute	0.32

Rebar

- Estimated by the ton
- Keeps type of placement (i.e., beams, columns, etc.) separate

Rebar Properties

Bar Designation	Nominal Diameter	Cross-sectional Area	Weight (pound per ft)
2	1/4″	0.015	0.167
3	3/8″	0.11	0.376
4	1/2″	0.20	0.668
5	5/8″	0.31	1.043
6	3/4″	0.44	1.502
7	7/8″	0.60	2.044
8	1″	0.79	2.670
9	1-1/8″	1.00	3.400
10	1-1/4″	1.27	4.303
11	1-3/8″	1.56	5.313

Rebar Splicing

Bar Size	Splice Based on Number of Bar Diameters	
	24 diameters	30 diameters
3	1′-0″	1′-0″
4	1′-0″	1′-3″
5	1′-3″	1′-7″
6	1′-6″	1′-11″
7	1′-9″	2′-3″
8	2′-0″	2′-6″
9	2′-4″	2′-10″
10	2′-7″	3′-3″
11	2′-10″	3′-7″

Wire Mesh

- W4x8 8/12
  - Longitudinal spacing = 4”
  - Transverse spacing = 8”
  - Longitudinal gauge = 8
  - Transverse gauge = 12

- Estimated by the roll or sheet

Reinforcing Productivity

Type of Placement	Productivity Rate (Labor Hrs/Ton)
Beams #3 to #7	22
Columns #3 to #7	24
Footings #4 to #7	15
Walls #3 to #7	11
Slab on Grade #3 to #7	13
Wire Mesh	1 per roll

Vapor Retarder

- Estimated by the roll

- Best to sketch a layout

Sq. Ft of Vapor Barrier	Labor Hour per 1,000 Sq. Ft.
Up to 1,000	1.5
1,000 to 5,000	1.25
Over 5,000	1

Accessories

- Expansion joint covers
- Expansion joint fillers
- Waterstop
- Manhole frames and covers
- Trench frames and covers

Types of Concrete Finishes

- Machine toweled
- Carborundum rubbing
- Wood float
- Broom
- Bush hammered
- Exposed aggregate
- Rubbed
- Sandblasted
- Ground
- Lightly sanded

Concrete Finishing

Finish	Method/Location	Labor Hrs/100 s.f.
Toweling	Machine - Slab on Grade	1.0
	Hand - Slab on Grade	2.0
Broom Finish	Slabs	0.75
	Stairs	4.0
Float Finish	Slab on Grade	3.5
	Slabs Suspended	4.0
	Walls	3.5
	Stairs	3.0
	Curbs	8.5

Concrete Finishing

Finish	Method/Location	Labor Hrs/100 s.f.
Bush Hammer	Machine – Green Concrete	3.0
	Machine – Cured Concrete	6.0
	Hand – Green Concrete	9.0
Rubbed	With Burlap and Grout	2.5
	With Float Finish	4.0

Curing

- Sealing compounds
- Heating in cold weather
- Wetting in warm weather

Concrete Transportation

- Truck chutes
- Buggies
- Crane and bucket
- Crane and hopper
- Tower and buggies
- Forklifts
- Conveyors
- Pumps

Forms

- Often bid by the square foot contact area
  - Be sure include:
    - Both sides
    - Ends
    - Bottom (if required)
- Cost is affected by number of uses

Form Productivity

Application	Productivity Rate (Labor Hours per SFCA)			
	1 use	2 use	3 use	4 use
Foundation Wall (Plywood)	0.44	0.28	0.25	0.22
Footings	0.14	0.09	0.08	0.07
Slab on Grade	1.2	1.1	1.0	0.8
Columns	0.180	0.146	0.143	0.140
Beams	0.2	0.18	0.15	0.14

Waffle Slab Form Liners

Void Size and Depth	Concrete Voided (c.f.)
19″x19″x4″	0.77
19″x19″x6″	1.09
19″x19″x8″	1.41
19″x19″x10″	1.9
19″x19″x12″	2.14
30″x30″x8″	3.85
30″x30″x10″	4.78
30″x30″x12″	5.53
30″x30″x14″	6.54
30″x30″x16″	7.44

Precast Concrete

- Double tee
- Single tee
- I girder
- Inverted T beam
- Monwing
- Box girder
- Hollow core slab
- Columns
- Plies
- Channel slab
- Wall panel

Estimating Precast

- Simply reinforce or prestressed?
- Coordination with:
  - Mechanical
  - Electrical
- Concrete fill (topping)
- Fire rating
- Welding

Thank You